1 | Literature Review

1.1 Ryan et.al.

Year: 2025

The QG^{+1} model incorporates the first-order corrections that were neglected in the basic QG approximation. It essentially refines the QG equations by accounting for non-geostrophic (ageostrophic) flow components that are dependent on the Rossby number (ϵ).

In **Chapter 2**, the QG^{+1} model is introduced. In this paper,

$$N = f \equiv \text{Constant}^{1}$$

[1] *page 8*

To facilitate the asymptotic approximation, a potential field is introduced.

$$\mathbf{A} = (-G, -F, \Phi)$$

By Incompressible condition we have

$$\mathbf{v} = \nabla_3 \times \mathbf{A}^{2}$$

[2] *In this paper ∇_3 is 3D gradient. 2D is just ∇*

Some Physical implications of the model

1. Breaking Symmetry of QG model.
2. It Captures Cyclogeostrophic balance.

Cyclogeostrophic balance is a fundamental force balance approximation used in meteorology and physical oceanography to describe the motion of fluids (like air and water) in curved paths, where the Coriolis force is balanced by the pressure gradient force and the centrifugal force. It is an essential extension of the simpler geostrophic balance, which only considers straight flow. This balance is particularly important in systems with high curvature and strong winds, such as tropical cyclones (hurricanes/typhoons), mid-latitude low-pressure systems, and strong ocean eddies. The Governing Equation is

$$\underbrace{f\mathbf{v}}_{\text{Coriolis Force}} + \underbrace{\frac{|\mathbf{v}|^2}{R}}_{\text{Centrifugal Force}} = \underbrace{-\frac{1}{\rho}\frac{\partial p}{\partial n}}_{\text{Pressure Gradient Force}} \tag{1.1}$$

Here n is the normal direction pointing toward the center of curvature.

3. Inclusiong of **Frontogenesis** [3].

[3] *Generation of Ocean Fronts*

In **Chapter 3.** A simulation for QG^{+1} is conducted, showing several features:

1. More Vigorous due to captureing ageostrophic frontogenesis.
2. Since the Ageostrophic effects creates stronger surface velocity. Finer structure can be seen on surface using QG^{+1}. [4].

[4] *See Figure 4 in page 24*

In summary, this paper provides a very detailed derivation to the QG^{+1} equation which is introduced more detailed in **??**. This paper also demonstrate two simulation to show how QG^{+1} model captures balanced submesoscale dynamics and frontogenesis.

Remark 1 In my own derivation following Ryan's work in the later chapter, first half of the calcualtion is based on this paper, then I follow Ryan's note for the rest.

1.1.1 Questions Regrading this Paper

1. In page 10, Equation 15. How does the above derivation implies $F = G = 0$. Does this implies that
$$F^0 + \epsilon F^1 + \cdots = 0$$
Then each order term of F and G is 0. I just don't get the physical meaning here.

1.2 J.Wang et.al. Reconstructing the Ocean's Interior from Surface Data

Year : 2013

In the **Introduction**, the author discussed the current challenge of using SSH and SST [5] measurement to reconstruct subsurface dynamics.

[5] *Surface Sea Height and Surface Sea Temperature*

- Traditional studies assume the signal is dominated by barotropic and first baroclinic modes. However, these modes are typically calculated by **assuming buoyancy anomalies vanish at the surface**.
- SQG theorys works as well. But it normaly assume 0 interior PV.

The author introduced the **Interior plus surface QG** method. It is quasigeostrophic. As introduced in Chapter 2:

1. Surface buoyancy anomoly contributes to the surface part of streamfunction ψ^s :
$$\mathcal{L}\Psi + f_0 + \beta y = Q,$$
$$\mathcal{L} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z}\frac{f_0^2}{N^2}\frac{\partial}{\partial z}\right), \quad \text{and } -H < z < 0,$$
Governing equation is
$$\mathcal{L}\psi^s = 0$$
Essentially, this is same in the SQG theory where we assume 0 interior PV. With boundary condition :
$$\frac{\partial}{\partial z}\psi^s(\mathbf{x}, z, t) = b(\mathbf{x}, z, t)/f_0 \quad \text{at} \quad z = 0, -H,$$
2. The interior part is governed by
$$\frac{\partial}{\partial z}\frac{f_0^2}{N^2}\frac{\partial}{\partial z}\hat{\psi}^i - \kappa^2\hat{\psi}^i = \hat{q}^i \quad \text{with} \quad \frac{d\hat{\psi}^i}{dz} = 0 \quad \text{at} \quad z = 0, -H.$$

However, the interior q^i is not clear.

Remark 2 This is the essential modification of this isQG model. They project the interior induced PV equation onto baroclinic modes and **impose additional boundary conditions to deduce the gravest modes.**

1.3 John R. Taylor and Andrew F. Thompson 2023

This review focuses on submesosclae dynamics in the upper ocean with a particular focus on multiscale interactions involving the submesoscale.

1.3.1 Generation of Submesoscale Flow

Submesosclaes are energized by a avariety of instabilities that develop from balanced currents.

1.3.2 Energy Transfer and Multiscale Interactions

Down-Scale Connections

Energy enters the ocean at large scales through surface wind. The mesocslae baroclinic instability then converts the energy to low vertical modes. Submesosclae motions are potential mechanisms for supplying energy from ocean mososcale to small-scale turbulence and scales that ultimately dissipated. [6]

Simulation shows that the formation of fronts is one of such path.

1.4 Ross et al. 2023

To coarse-grain a high-resolution field $\hat{q}_\kappa$ onto a low-resolution grid, Ross et al. propose three filtering and coarse-graining operators. Let κ denote the wavenumber, Δx_{lores} the grid spacing of the coarse model, and define the cutoff $\kappa^c \equiv 2/3$ of the low-resolution Nyquist frequency.

1.4.1 Operator 1: Spectral Truncation with Sharp Filter

The first option applies the same quadruple-exponential filter used by pyqg for small-scale dissipation. It leaves small wavenumbers unchanged and attenuates wavenumbers above κ^c:

$$\hat{\bar{q}}_\kappa = \begin{cases} \hat{q}_\kappa, & \kappa < \kappa^c, \\ \hat{q}_\kappa \cdot e^{-23.6(\kappa-\kappa^c)^4 \Delta x_{\text{lores}}^4}, & \kappa \geq \kappa^c. \end{cases} \tag{1.2}$$

This is the most conservative choice, closest to no filtering at all, since it is already applied within the ocean model. [7]

1.4.2 Operator 2: Spectral Truncation with Softer Gaussian Filter

The second option applies a Gaussian filter to all remaining modes:

$$\hat{\bar{q}}_\kappa = \hat{q}_\kappa \cdot e^{-\kappa^2 (2\Delta x_{\text{lores}})^2/24}. \tag{1.3}$$

This choice follows Guan et al. (2022) and Pope (2000). By the filter-width definition of Lund (1997), this filter is twice as large as the grid size of the coarse model.

1.4.3 Operator 3: Diffusion-Based Filtering with Real-Space Coarsening

The third option is closer to the procedure used in ocean models. It applies `GCM-Filters` (Grooms et al. 2021; Loose et al. 2022), which approximates the spectral transfer function of a Gaussian filter using polynomials based on the Laplacian diffusion operator. The pyqg output is converted onto a Cartesian grid, and the filtered field is then coarse-grained in real space: to reduce the resolution by a factor K, the input field is averaged over non-overlapping $K \times K$ boxes.

Remark 3

> A comparison of the effects of these three filtering and coarse-graining operators on PV and its subgrid forcing is shown in Figures 3 and 4 of Ross et al. (2023).